

The Program in Action

When you run *periodictable.py*, the output will look like this:

Periodic Table of Elements

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```
Periodic Table of Elements
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18
1 H                               He
2 Li Be                         B  C  N  O  F  Ne
3 Na Mg                         Al Si P  S  Cl Ar
4 K  Ca Sc Ti V  Cr Mn Fe Co Ni Cu Zn Ga Ge As Se Br Kr
5 Rb Sr Y  Zr Nb Mo Tc Ru Rh Pd Ag Cd In Sn Sb Te I  Xe
6 Cs Ba La Hf Ta W  Re Os Ir Pt Au Hg Tl Pb Bi Po At Rn
7 Fr Ra Ac Rf Db Sg Bh Hs Mt Ds Rg Cn Nh Fl Mc Lv Ts Og
```

```
Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu
```

```
Th Pa U  Np Pu Am Cm Bk Cf Es Fm Md No Lr
```

Enter a symbol or atomic number to examine, or QUIT to quit.

> 42

Atomic Number: 42

Symbol: Mo

Element: Molybdenum

Origin of name: Greek molýbdaina, 'piece of lead', from mólybdos, 'lead'

Group: 6

Period: 5

Atomic weight: 95.95(1) u

Density: 10.22 g/cm^3

Melting point: 2896 K

Boiling point: 4912 K

Specific heat capacity: 0.251 J/(g*K)

Electronegativity: 2.16

Abundance in earth's crust: 1.2 mg/kg

Press Enter to continue...

--snip--

How It Works

A *.csv*, or *comma-separated values*, file is a text file that represents a primitive spreadsheet. Each line in the *.csv* file is a row, and commas separate the columns. For example, the first three lines in *periodictable.csv* look like this:

```
1,H,Hydrogen,"Greek elements hydro- and -gen, meaning 'water-forming"--snip--
2,He,Helium,"Greek hēlios, 'sun'",18,1,4.002602(2)[III][V],0.0001785--snip--
3,Li,Lithium,"Greek líthos, 'stone'",1,2,6.94[III][IV][V][VIII][VI],--snip--
```

Python's *csv* module makes it easy to import data from a *.csv* file and into a list of lists of strings, as lines 15 to 18 do. Lines 32 to 58 turn this list of lists into a dictionary so that the rest of the program can easily summon the information by an element's name or atomic number.